

Blackouts Preserving Rectangle Identifiability on Grid Points

A partial solution to Problem 001 of Kagey's Open Problem Collection

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Abstract

We study Problem 001 of Kagey's Open Problem Collection: black out as many points of an $n \times m$ grid as possible so that every possible rectangle drawn with its corners on grid points remains uniquely identifiable from its visible corners. We reduce the problem to a minimum hitting set of a difference hypergraph, solve the two-row case completely (the Strip Theorem: maximum blackout $m + 1$ with exactly $m \cdot 2^{m-1}$ optima for $m \geq 3$), prove that the maximum blackout is $m + 2$ on three rows for $m \geq 3$ and $m + 4$ on four rows for $m \geq 6$, and obtain the general upper bound $m + \lfloor n^2/4 \rfloor$ for $m \geq 3$. The structural step characterizes axis-aligned identification by the absence of four- and six-cycles in the black row-column incidence graph, and a separate geometric argument handles tilted rectangles. We also report the computations, the uniqueness of the 5×5 optimum up to symmetry (by computer), and the remaining conjecture for larger grids.

1 Introduction

The problem is posed with some alterations chosen for clarity as the following: Suppose you, the player, are shown an $n \times m$ grid of vertices. Each vertex has two states, 'white' or 'blackened out'. A vertex that is blackened out is unable to convey any information. The game master asks us to 'black out', or turn off as many of the vertices as we like. Once we choose the 'blackout', which is simply the set of all vertices that are blackened out, the game master will draw a rectangle that we cannot see but know the corners of given that the corners are on a white vertex. The rectangle's sides are not necessarily parallel to the grid axes. We must be able to guess what the rectangle's corners are from this information with complete confidence. The question posed by Kagey [1] is how many vertices can be blackened out while every rectangle the game master might draw remains distinctly identifiable.

The order of play is important here. Because the blackout is chosen by us before the rectangle is chosen, a blackout is valid only if *no* two rectangles are confusable under it. Writing $C(R)$ for the corner set of a rectangle and S for the blackout, what the player sees is $C(R) \setminus S$; the requirement is that this assignment be injective on the set of rectangles. Section 2 makes this precise, and the rest of the paper studies the resulting extremal quantity.

The Strip Theorem (Theorem 1) settles two rows, including the classification and number of optima. Theorems 3 and 4 settle the values on three rows and on four rows of width at least 6. The four-row grids of width 4 and 5 are settled by an exhaustive computer check (Corollary 2). The general row bound and the short-cycle characterization are Theorem 2 and Lemma 2. Optimum

*Code and data: github.com/apemm/Kagey-Problems.

counts beyond strips and the 5×5 uniqueness statement remain computational. The higher-dimensional Excess-1 formula remains conjectural (Conjecture 1). The old guess that the kept set always has size $(n-1)(m-1)$ was first refuted at 5×5 and also fails on $4 \times m$ for every $m \geq 6$.

Every prediction was written in a ledger before the computation that tested it. The values were computed with an ILP formulation, a branch-and-bound oracle, and an exhaustive check straight from the definition, and Table 1 says which of these agree for each grid. The three- and four-row results have their own verifier, which enumerates the rectangles by their diagonals.

The paper is organized as follows. Section 2 fixes conventions and defines the presentation map. Section 3 proves the reduction and its consequences. Section 4 describes the pruning of the difference hypergraph. Section 5 proves the Strip Theorem and Section 6 proves the row bound and the results for three and four rows. Section 7 reports the computations and the verification. Section 8 treats uniqueness at 5×5 . Section 9 records the refuted and live conjectures and states open questions.

2 Conventions and the presentation map

Vertices of the $n \times m$ grid are denoted as pairs (i, j) with $1 \leq i \leq m$ (column) and $1 \leq j \leq n$ (row). The set of all vertices on the grid is V . The rectangle class consists of all rectangles with four corners on grid points, tilted rectangles included and degenerate (zero-area) configurations excluded; $C(R)$ denotes the corner set of a rectangle R . For a blackout $S \subseteq V$ the *presentation map* sends each rectangle to its visible corners, $P_S(R) = C(R) \setminus S$, and S is *valid* when P_S is injective over all rectangles. The empty presentation is an ordinary and accepted value of the map, that at most one rectangle may be fully hidden is a corollary of injectivity, not an axiom (Remark 1).

Remark 1. Suppose 2 distinct rectangles $R_1 \neq R_2$ with corner sets C_1 and C_2 are both fully hidden, that is, $C_1 \subseteq S$ and $C_2 \subseteq S$. Then $P_S(R_1) = \emptyset = P_S(R_2)$, and since $\emptyset$ is an ordinary value of the presentation map, injectivity gives $R_1 = R_2$, contradicting our supposition. Hence, at most 1 rectangle may be fully hidden under any valid blackout, with no separate convention required.

3 The reduction

Lemma 1 (difference-set characterization). *For any sets A, B, S : $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$.*

Proof. ($\Rightarrow$) Assume $A \setminus S = B \setminus S$, and let $x \in A \triangle B$ be arbitrary. Exchanging A and B makes no difference since $\triangle$ is commutative and symmetric about A and B , so we may assume $x \in A \setminus B$. Suppose $x \notin S$. Then $x \in A \setminus S$, and since $A \setminus S = B \setminus S$ we get $x \in B \setminus S$, hence $x \in B$, contradicting $x \in A \setminus B$. Therefore $x \in S$. As x was arbitrary, $A \triangle B \subseteq S$.

($\Leftarrow$) Assume $A \triangle B \subseteq S$. Both the hypothesis and the conclusion are unchanged when A and B are exchanged, so it suffices to prove $A \setminus S \subseteq B \setminus S$. Let $x \in A \setminus S$ be arbitrary, so $x \in A$ and $x \notin S$. Suppose $x \notin B$. Then $x \in A \setminus B \subseteq A \triangle B \subseteq S$, contradicting $x \notin S$. Hence $x \in B$, and with $x \notin S$ we get $x \in B \setminus S$. As x was arbitrary, $A \setminus S \subseteq B \setminus S$; the reverse inclusion follows by symmetry, so $A \setminus S = B \setminus S$. $\square$

Corollary 1. *S is valid if and only if the kept set $K = V \setminus S$ intersects the difference set $C(R_1) \triangle C(R_2)$ for every pair of distinct rectangles. Hence a maximum blackout is the complement of a minimum hitting set of the difference hypergraph.*

Proof. By definition, S is valid if and only if $P_S(R_1) \neq P_S(R_2)$, i.e. $C(R_1) \setminus S \neq C(R_2) \setminus S$, for every pair of distinct rectangles. From Lemma 1, this holds if and only if $C(R_1) \triangle C(R_2) \not\subseteq S$, which is exactly the statement that at least 1 point of the difference set lies in K . $\square$

4 Preprocessing

Prior to any computation, we prune the hypergraph for convenience and clarity. The first technique is deduplication, since identical difference sets impose identical constraints on the allowed K , so we only need to keep one copy. The second technique is a containment rule: if D_1 and D_2 are two difference sets and $D_1 \subset D_2$, any K intersecting D_1 automatically intersects D_2 as well, so D_2 can be omitted from the hypergraph without any impact on which K are valid. Third, an empty row remark: an empty difference set would mean that $C_1 \triangle C_2 = \emptyset$, hence $C_1 = C_2$, no K can intersect this difference set, so its presence would signal that the rectangle list was not deduplicated rather than a genuine constraint.

With these pruning techniques, the hypergraph can be replaced by its pruned antichain of minimal rows. All of the following computations in Section 7 run on the pruned version of the hypergraph.

5 The Strip Theorem

Theorem 1 (Strip Theorem). *For a $2 \times m$ grid with $m \geq 3$, every maximum valid blackout has size $m + 1$, there are exactly $m \cdot 2^{m-1}$ of them, and they are characterized as the complements of kept sets containing one survivor per column and missing exactly one column. For $m = 2$ the condition is vacuous and every blackout is valid.*

Proof. Define K as the visible set, where $V \setminus S = K$, and let K_0 denote the minimum possible size of K over valid blackouts. For a $2 \times m$ grid, define the set T to be the set of all columns with at least 1 of their 2 points in K . For $m = 2$ there is exactly one rectangle, so injectivity is vacuous and every blackout is valid; assume $m \geq 3$. Suppose S is a maximum valid blackout.

On a $2 \times m$ strip every rectangle is axis-aligned, since a tilted rectangle's corners would occupy at least 3 distinct rows and the strip has only 2. We can represent any possible rectangle R as $R_{i,j}$, where i is the column containing the left corners of R and j is the column containing the right corners, $1 \leq i < j \leq m$.

If two distinct rectangles share exactly one column, say $R_{z,p}$ and $R_{z,q}$, the shared column's corners overlap, and $C(R_{z,p}) \triangle C(R_{z,q})$ is exactly the 4 points of columns p and q ; we call this the pair row on $\{p, q\}$. If they share no column, the difference set is all 8 points of 4 distinct columns, which contains a pair row and is deleted in preprocessing (Section 4). Every pair $\{p, q\}$ arises this way: two rectangles sharing a column span at least 3 distinct columns between them, and $m \geq 3$ guarantees a third column z from which to build $R_{z,p}$ and $R_{z,q}$. Hence, the pruned hypergraph consists of exactly the $\binom{m}{2}$ pair rows.

Then, from Lemma 1, a valid visible set K is one that intersects every pair row. Conditioning on T , this constrains T to contain every column except at most one: if two distinct columns p and q are excluded, the pair row on $\{p, q\}$ contains no point of K , and K would not intersect every pair row. Each column of T contributes at least 1 point, so $|K| \geq |T| \geq m - 1$, and hence $K_0 = m - 1$ and $|S| \leq 2m - (m - 1) = m + 1$. If no column is fully blacked out, T would contain every column and K would need m points, so $|S| \leq m$. If any column of T kept both of its points, $|K| \geq m$ as well. Hence, at the maximum, exactly one column is excluded from T , and in the other $m - 1$ columns, exactly 1 of the 2 points in each column is blacked out.

Every such construction is valid: any pair of columns includes at least one column of T , whose surviving point meets the pair row. There are m ways to choose which column to exclude from T and 2^{m-1} ways to choose which 1 of the 2 points to black out in the remaining $m-1$ columns. Hence, $m \cdot 2^{m-1}$ such distinct maximum blackouts exist, characterized as in the statement. $\square$

6 Three and four rows

In this section coordinates start at zero: columns are $0, \dots, m-1$ and rows are $0, \dots, n-1$.

Lemma 2 (axis-aligned identification and short cycles). *Suppose $m \geq 3$. Form a bipartite graph G_S whose vertices are the rows and columns, with an edge between row y and column x exactly when $(x, y) \in S$. The axis-aligned rectangles have distinct presentations if and only if G_S has neither a cycle of length four nor a cycle of length six.*

Proof. If rows a, b are black in both columns x, z , choose a third column w . The rectangles on these rows and column pairs $\{x, w\}$ and $\{z, w\}$ have the same presentation. Thus a four-cycle is forbidden. If a six-cycle has successive vertices a, x, b, y, c, z, a , the rectangles with row/column pairs $(\{a, b\}, \{x, y\})$ and $(\{a, c\}, \{y, z\})$ have every corner black except possibly their common corner (y, a) . Their presentations coincide.

Conversely, assume both cycles are absent. No rectangle is entirely black, since its four edges would form a four-cycle. Two rectangles with a common presentation of at least three points coincide, since three rectangle corners determine the fourth. If two visible points are opposite axis-aligned corners, they determine the axis-aligned rectangle. If they lie in one row or column, two distinct completions require a black four-cycle at the two other rows or columns. Finally, suppose the common presentation is the single corner (x, a) . Write the other row and column of the two rectangles as (b, y) and (c, z) . If $b = c$ or $y = z$, their black corners contain a four-cycle. Otherwise the six black edges form the cycle a, y, b, x, c, z, a . Every possibility contradicts the assumption. $\square$

Theorem 2 (general row bound). *For $m \geq 3$, every valid blackout on an $n \times m$ grid satisfies*

$$|S| \leq m + \left\lfloor \frac{n^2}{4} \right\rfloor.$$

If also $n \geq 3$, the transposed bound holds as well.

Proof. For each nonempty column let B_x be its set of black rows, and choose any tree on B_x (no edges if $|B_x| = 1$). Superimpose these trees into a graph H on the n row vertices. It is simple: repeating an edge would mean two columns share two black rows, a four-cycle in G_S . It is triangle-free. A triangle cannot lie within one tree; if two edges come from one column and the third from another, those columns share two rows; if the three edges come from three columns, they give a six-cycle in G_S . All cases are forbidden by Lemma 2.

Recall that a triangle-free graph on n vertices with $e > 0$ edges satisfies $d(u) + d(v) \leq n$ for every edge uv . Summing over edges and using Cauchy–Schwarz gives

$$\frac{4e^2}{n} \leq \sum_v d(v)^2 \leq ne,$$

so $e \leq \lfloor n^2/4 \rfloor$, the usual Mantel bound. If $t \leq m$ columns are nonempty, the construction gives $e = \sum_{B_x \neq \emptyset} (|B_x| - 1) = |S| - t$. Therefore $|S| = t + e \leq m + \lfloor n^2/4 \rfloor$. $\square$

Lemma 3 (tilted rectangles on narrow grids). *Every tilted rectangle on three rows is a translate of*

$$D(x, y) = \{(x, y+1), (x+1, y), (x+2, y+1), (x+1, y+2)\}, \quad y = 0.$$

On four rows, these diamonds have $y = 0, 1$, and the remaining tilted rectangles are the following four types, for $0 \leq x \leq m-4$:

$$\begin{aligned} A_x &= \{(x, 2), (x+1, 3), (x+2, 0), (x+3, 1)\}, \\ B_x &= \{(x, 2), (x+2, 3), (x+1, 0), (x+3, 1)\}, \\ C_x &= \{(x, 1), (x+1, 3), (x+2, 0), (x+3, 2)\}, \\ E_x &= \{(x, 1), (x+2, 3), (x+1, 0), (x+3, 2)\}. \end{aligned}$$

Proof. At the unique leftmost corner, write the side vectors as (a, b) and $(c, -d)$, where a, b, c, d are positive integers. Orthogonality says $ac = bd$, and the vertical span is $b + d$. If this span is at most two, all four integers are one. If it is three, $\{b, d\} = \{1, 2\}$ and $\{a, c\} = \{1, 2\}$, giving the four listed cases. These exhaust the possibilities. $\square$

Theorem 3 (Three-Row Theorem). *For every $m \geq 3$, the maximum valid blackout on a $3 \times m$ grid is $m + 2$.*

Proof. The upper bound is Theorem 2. For the lower bound take

$$S = \{(x, 0) : 0 \leq x < m\} \cup \{(0, 1), (0, 2)\}.$$

Its incidence graph is a tree, so its axis-aligned presentations are distinct. By Lemma 3, each tilted rectangle is a diamond $D(x, 0)$. For $x > 0$ it has only one black corner, hence at least three visible corners and cannot be confused with any other rectangle. For $x = 0$ its presentation is $\{(2, 1), (1, 2)\}$. The unique axis-aligned rectangle with these opposite corners also contains the white points $(1, 1), (2, 2)$, so cannot have this presentation. There is no second tilted rectangle with at most two visible corners. Thus S is valid and has $m + 2$ points. $\square$

Theorem 4 (Four-Row Theorem). *For every $m \geq 6$, the maximum valid blackout on a $4 \times m$ grid is $m + 4$.*

Proof. The upper bound is Theorem 2. For the matching construction set

$$\begin{aligned} S &= \{(x, 0) : 1 \leq x \leq m-2\} \\ &\cup \{(m-2, 1), (m-1, 1), (0, 2), (1, 2), (0, 3), (m-1, 3)\}. \end{aligned}$$

Its size is $m + 4$. Its columns with two black points are $0, 1, m-2, m-1$, joining row pairs $\{2, 3\}, \{0, 2\}, \{0, 1\}, \{1, 3\}$, respectively. Thus G_S is an eight-cycle with pendant edges, and all axis-aligned presentations are distinct.

It remains to check tilted rectangles. Any rectangle with at least three visible corners is uniquely determined, so only rectangles with at least two black corners need attention. All diamonds $D(x, 1)$, and all C_x, E_x , have at most one black corner. Put $L = \{0, 1\}$ and $R = \{m-5, m-4\}$. The remaining presentations with at most two visible corners are exactly those in the table. In the last three rows $F = A$ or B , with $s = 1$ or 2 , respectively.

rectangle	range	visible corners
$D(0, 0)$		$\{(0, 1), (2, 1)\}$
$D(x, 0)$	$x = m-4, m-3$	$\{(x, 1), (x+1, 2)\}$
F_x	$x \in L \setminus R$	$\{(x+s, 3), (x+3, 1)\}$
F_x	$x \in R \setminus L$	$\{(x, 2), (x+s, 3)\}$
F_x	$x \in L \cap R$	$\{(x+s, 3)\}$

The table follows by checking membership in S . Its two-point presentations are distinct: their row pairs and horizontal separations distinguish the types, and within a type the coordinates determine x . The only one-point presentations occur at $m = 6, x = 1$, and are $(2, 3)$ and $(3, 3)$, which are distinct.

None equals an axis-aligned presentation. For $\{(0, 1), (2, 1)\}$ no other row is black in both columns 0, 2. For $\{(x, 1), (x + 1, 2)\}$ the other axis-aligned corner $(x, 2)$ is white, since $x \geq 2$. In the $L \setminus R$ case, $(x + s, 1)$ is white, since $x + s \leq 3 < m - 2$. In the $R \setminus L$ case, $(x, 3)$ is white, since $0 < x < m - 1$. Finally, when $m = 6$, an axis-aligned rectangle with sole visible corner $(z, 3)$, $z = 2$ or 3 , would need another column $c = 0$ or 5 and another row black in both columns z, c . No such row exists: row zero is white at 0, 5, and rows one and two are white at z . Hence S is valid. $\square$

Lemma 4 (four-row upper-bound equality cases). *If $m \geq 4$ and a valid blackout has $m + 4$ black points on four rows, then every column is nonempty, exactly four columns have two black points, the others have one, and the four double columns form a four-cycle on the row vertices.*

Proof. Equality in Theorem 2 requires every column nonempty and total excess $\sum_x (|B_x| - 1) = 4$. A column containing all four rows leaves all other columns with at most one black point, giving excess three. If a column contains three rows, any other nonsingleton column must pair the fourth row with one of those three. There can be at most one such column: two with the same row pair give a four-cycle in G_S , and two with different row pairs give a six-cycle. The total excess is then at most $2 + 1 = 3$. Thus all nonsingleton columns have size two. They give a simple triangle-free graph with four vertices and four edges, necessarily a four-cycle. $\square$

Corollary 2 (finite boundary cases, computer-assisted). *The maximum blackout is 7 on 4×4 and 8 on 4×5 .*

Proof. The upper bound leaves only a possible value $m + 4$ to exclude. By Lemma 4, candidates are generated by choosing one of the three four-cycles on the row labels, four columns, a bijection from their four row-pair edges to those columns, and a row for each remaining singleton column. This is the complete list of

$$3 \cdot 4! \binom{m}{4} 4^{m-4}$$

candidates: 72 when $m = 4$ and 1440 when $m = 5$. The standard-library script `verify_narrow.py` checks every candidate directly against all rectangles, generated independently from equal-length diagonals with a common midpoint. Every candidate has a repeated presentation. The same script checks that the blackout consisting of row zero and column zero is valid in these two grids, attaining $m + 3$. $\square$

The verifier also checks Lemma 2 on all 74,304 blackouts of five small grids, compares the lists of tilted rectangles in Lemma 3 with the diagonal enumeration, and checks the constructions for widths 3, ..., 40, 64, 100, 128 where they apply. These checks are only a test of the proofs above; the proofs do not depend on them.

7 Computations and verification

The last column says which methods agree on each value. The values for three and four rows also follow from Section 6, but the counts of optima are only computational. Rectangle counts match the literature: $n \times n$ counts 1, 10, 44, 130 agree with OEIS A085582, and $3 \times m$ counts with OEIS A289832 (indexing note in the bibliography).

grid	rects	blackout	kept	optima	verification
2×2	1	4	0	1	three-way
2×3	3	4	2	12	three-way
2×4	6	5	3	32	three-way
2×5	10	6	4	80	three-way; pre-registered
2×6	15	7	5	192	engine + oracle
2×7	21	8	6	448	engine + oracle
3×3	10	5	4	53	three-way
3×4	20	6	6	224	three-way
3×5	33	7	8	892	three-way
3×6	49	8	10	3420	three-way
4×4	44	7	9	316	three-way
4×5	74	8	12	1810	three-way
3×7	68	9	12	12704	engine + brute
3×8	90	10	14	45864	engine + brute
3×9	115	11	16	161992	engine + brute
3×10	143	12	18	—	engine only (value)
4×6	110	10	14	26	engine + brute
4×7	152	11	17	292	engine + brute
4×8	200	12	20	—	engine only (value)
5×5	130	10	15	8	engine + brute
5×6	198	11	19	—	engine only (value)
5×7	276	12	23	—	engine only (value)
6×6	313	12	24	—	engine only (value)

Table 1: Exact values. “Three-way” = engine, oracle, and definitional brute agree; “engine + brute” = the ILP engine and the exhaustive definitional check agree; “engine only” = the exhaustive check is infeasible at that size.

8 Uniqueness at 5×5

Theorem 5 (computational). *The 5×5 grid has maximum blackout 10, attained by exactly 8 blackouts, and the maximum blackout is unique up to the symmetries of the square: the 8 optima form a single orbit of the dihedral group D_4 , with trivial stabilizer.*

The value and the count were produced independently by the engine and by the exhaustive definitional check; the orbit claim was verified by applying the 8 symmetries of the grid to one optimum and recovering all 8. The unique extremal blackout is the 9-point anti-diagonal staircase, the old $n + m - 1$ pattern, plus 1 detached corner: the +1 made visible. A representative (‘#’ blacked out, ‘.’ kept):

```

. . . . .
. . # . .
. # . . .
# # . . .
# . . . #

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The kept set is two triangular blocks of 5 and 10 points on either side of the band, and the extremal object is unique up to symmetry while having no symmetry of its own.

9 Refuted and live conjectures

conjecture	status	killed by
$\text{kept} = (n - 1)(m - 1)$	refuted (pre-registered)	5×5
$H_{\text{sum}}: \text{blackout} = n + m - 1 + \lceil n + m \geq 10 \rceil$	refuted (pre-registered)	3×7
$H_{\text{area}}: \text{blackout} = n + m - 1 + \lceil nm \geq 24 \rceil$	refuted (pre-registered)	3×8
Four-Row: $\text{kept}(4, m) = \lceil 5m/2 \rceil - 1$	refuted (pre-registered)	4×8

The Three-Row conjecture, which we checked for $3 \leq m \leq 10$ before proving it, is now Theorem 3. The four-row case of Excess-1 is Theorem 4, with the boundary cases in Corollary 2. We have not classified or counted the optima beyond two rows.

Conjecture 1 (Excess-1). $\text{blackout}(n, m) = n + m$ for $\min(n, m) \geq 4$, *excepting* 4×4 and 4×5 up to transposition, where it is $n + m - 1$.

We checked this on seven cells (4×6 through 6×6), including the prediction of 12 at 6×6 , which was written down before the 1063-second computation. The four-row cases are now proved, and the case where both dimensions are at least 5 is open. The row bound does not imply Excess-1 there.

Beyond the remaining conjecture, three questions strike us as the natural next ones: why the number of optima collapses at 4×6 (26, against neighbors in the hundreds and thousands); whether the excess ever grows past 1, with 5×8 and 6×7 registered at 13 as the next kill attempts; and whether the 5×5 uniqueness admits a structural proof rather than a computational one.

Data and code availability

The code, tables, and ledger are at <https://github.com/apemm/Kagey-Problems>. The two four-row boundary cases are computer-assisted.

Acknowledgment

The Lean code for this paper was generated by AI. AI was also used to revise some of the writing and to help cross-verify the code.

References

- [1] P. Kagey, Problem 001, Open Problem Collection, peterkagey.com/problems/001/ (accessed August 2026).
- [2] OEIS Foundation, Sequence A085582: number of rectangles (orthogonal or not) with corners on an $n \times n$ grid of points. <https://oeis.org/A085582>. Verified against Table 1: 1, 10, 44, 130 at $n = 2..5$.
- [3] OEIS Foundation, Sequence A289832: table of rectangle counts on rectangular grids; diagonal equals A085582. <https://oeis.org/A289832>. Indexing: $T(n, k)$ counts rectangles on an $(n+1) \times (k+1)$ point grid, so an $r \times c$ point grid appears as $T(c-1, r-1)$; e.g. our 3×7 count is $T(6, 2) = 68$.